Supporting Information. Ocampo, D., K.G. Borja-Acosta, J. Lozano-Flórez, S. Cifuentes-Acevedo, E. Arbeláez-Cortés, N.J. Bayly, Á. Caguazango, B. Coral-Jaramillo, D. Cueva, F. Forero, J.P. Gómez, C. Gómez, M.A. Loaiza-Muñoz, G.A. Londoño[9], S. Pérez-Peña, S. Losada-Prado, H.E. Ramírez-Chaves, M.E. Rodríguez-Posada, J. Sanabria-Mejía, M. Sánchez-Martínez, V.H. Serrano-Cardozo[13], M. del Socorro Sierra-Buitrago, J. Soto-Patiño, and O. Acevedo-Charry. 2021. Body mass data set for 1317 bird and 270 mammal species from Colombia. Ecology

Metadata

Class I. Data set description

A. Data set identity

Title: Body mass data set for 1317 bird and 270 mammal species from Colombia

Data set identification code

DataS1_Body mass by species_DOetal.csv

DataS1_Body mass records_DOetal.csv

B. Data set description

1. Originators

David Ocampo[1], Kevin G. Borja-Acosta[1], Julián Lozano-Flórez[1], Sebastián Cifuentes-Acevedo[1,2], Enrique Arbeláez-Cortés[3], Nicholas J. Bayly[4], Ángela Caguazango[4], Brayan Coral-Jaramillo[5], Diego Cueva[1], Fernando Forero[1,6], Juan P. Gómez[7], Camila Gómez[4,8], Mario A. Loaiza-Muñoz[9], Gustavo A. Londoño[9], Sebastián Pérez-Peña[1], Sergio Losada-Prado[10], Héctor E. Ramírez-Chaves[11], Miguel E. Rodríguez-Posada[12], Jeyson Sanabria-Mejía[4], Manuel Sánchez-Martínez[9], Victor Hugo Serrano-Cardozo[13], M. del Socorro Sierra-Buitrago[1], Juliana Soto-Patiño[1] & Orlando Acevedo-Charry[1,14]

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2. Abstract

Body mass is one of the most important phenotypic attributes in animal ecology and life history. This trait is widely used in the fields of ecology and macroevolution, since it influences physiology, morphological functions, as well as a myriad of ecological and social interactions. In this data set, our aim was to gather a comprehensive bird and mammal body mass data set from northern South America. We report body mass, discriminated by sex, for 42,022 individual birds and 7,441 mammals representing 1317 bird species (69% of Colombia's avifauna) and 270 mammal species (51% of Colombian mammals) from the Neotropics. The data was sourced from vouchers collected between 1942 and 2020 and from individuals captured and released at banding stations over the last two decades for birds (2000 – 2020) and the last decade for mammals (2010 – 2020), by ten research groups and institutions in Colombia. This data set fills gaps identified in other similar databases, as it focuses on northern South America, a highly diverse Neotropical region often underrepresented in morphological data sets. We provide wide taxonomic coverage for studies interested in information both at regional and local scales. There are no copyright restrictions; the present data paper should be appropriately

cited when data are used for publication. The authors would appreciate learning about research projects, teaching exercises and other uses given to this data set and are open to contribute in further collaborations using this data.

3. Keywords

Aves; capture-release; Colombia; Mammalia; morphological trait; museum specimens; scientific collections; banding stations.

Class II. Research origin description

A. Overall project description

1. Identity

Body mass data set for 1317 bird and 270 mammal species from Colombia

2. Originators

David Ocampo[1], Kevin G. Borja-Acosta[1], Julián Lozano-Flórez[1], Sebastián Cifuentes-Acevedo[1,2], Enrique Arbeláez-Cortés[3], Nicholas J. Bayly[4], Ángela Caguazango[4], Brayan Coral-Jaramillo[5], Diego Cueva[1], Fernando Forero[1,6], Juan P. Gómez[7], Camila Gómez[4,8], Mario A. Loaiza-Muñoz[9], Gustavo A. Londoño[9], Sebastián Pérez-Peña[1], Sergio Losada-Prado[10], Héctor E. Ramírez-Chaves[11], Miguel E. Rodríguez-Posada[12], Jeyson Sanabria-Mejía[4], Manuel Sánchez-Martínez[9], Victor Hugo Serrano-Cardozo[13], M. del Socorro Sierra-Buitrago[1], Juliana Soto-Patiño[1] & Orlando Acevedo-Charry[1,14]

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[13]*Colección de Mastozoología y Laboratorio de Ecología, Grupo de Estudios en Biodiversidad, Universidad Industrial de Santander, Bucaramanga, Santander, Colombia*
[14]*Asociación Colombiana de Ornitología*, *Bogotá, D.C., Colombia*

3. Period of study

Body mass data set for 1317 bird and 270 mammal species from a compilation of scientific specimens collected between 1942 and 2020 (Figure 1). The collection data set was complemented using capture-release data gathered by the authors through the last two decades for birds (2000 – 2020) and the last decade for mammals (2010 – 2020).

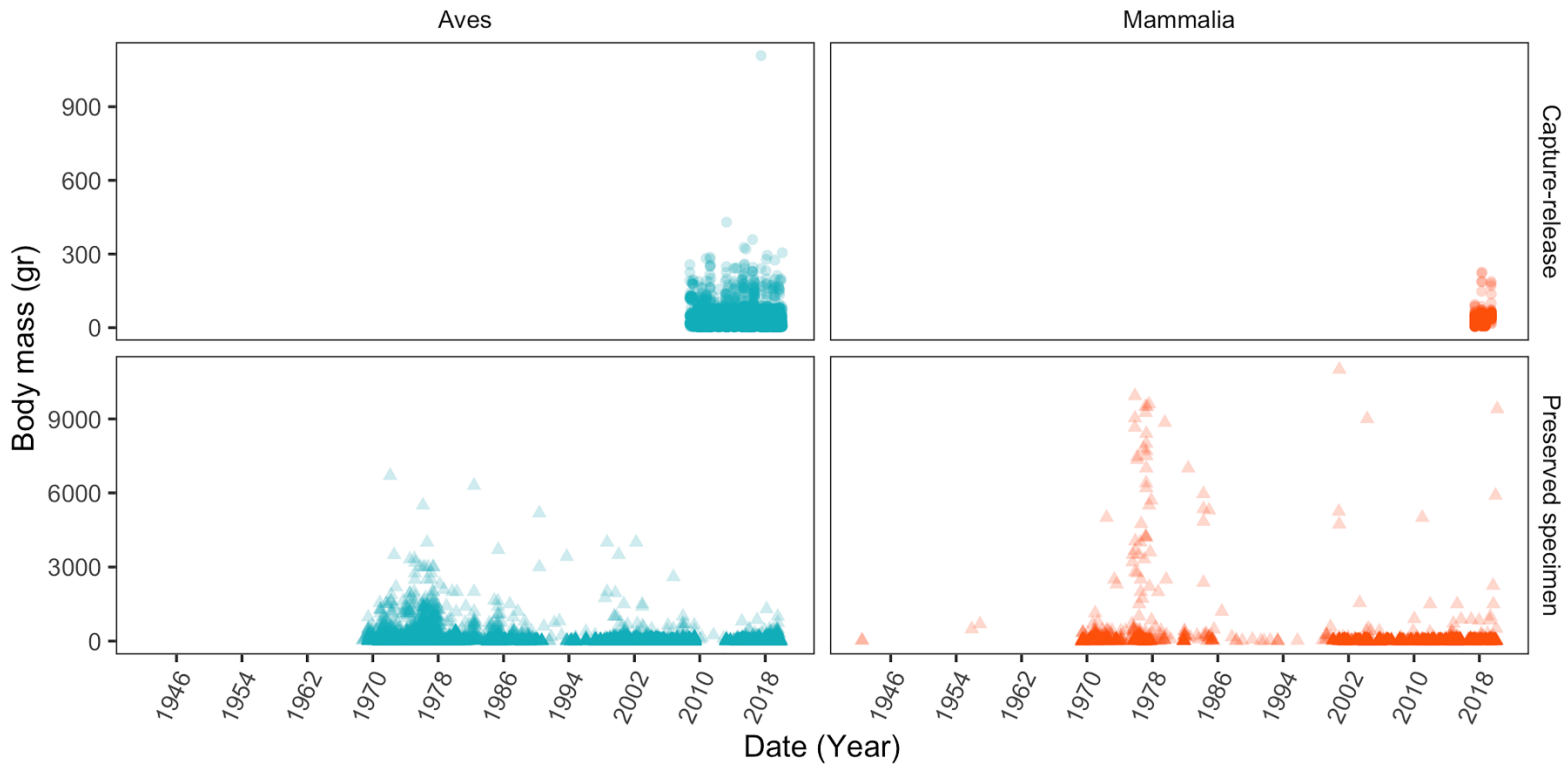

Fig. 1. Temporal distribution of birds and mammal body masses collected in Colombia between 1942 and 2020. Note that the heaviest specimens for both groups were more commonly collected during the 70's and 80s'.

4. Objectives

Our objective was to compile a comprehensive bird and mammal body mass data set from northern South America, with the aim of filling data gaps in public data sets for some taxa (Dunning 2007, Willman *et al*. 2014), focusing on South American species found in Colombia. As such, the data set covers a wide taxonomic range for both global and local studies in ecology and evolutionary biology.

5. Abstract

Body mass is one of the most important phenotypic attributes in animal ecology and life history. This trait is widely used in the fields of ecology and macroevolution, since it influences physiology, morphological functions, as well as a myriad of ecological and social interactions. In this data set, our aim was to gather a comprehensive bird and mammal body mass data set from northern South America. We report body mass, discriminated by sex, for 42,022 individual birds and 7,441 mammals representing 1317 bird species (69% of Colombia's avifauna) and 270 mammal species (51% of Colombian mammals) from the Neotropics. The data was sourced from vouchers collected between 1942 and 2020 and from individuals captured and released at banding stations over the last two decades for birds (2000 – 2020) and the last decade for mammals (2010 – 2020), by ten research groups and institutions in Colombia. This data set fills gaps identified in other similar databases, as it focuses on northern South America, a highly diverse Neotropical region often underrepresented in morphological data sets. We provide wide taxonomic coverage for studies interested in information both at regional and local scales. There are no copyright restrictions; the present data paper should be appropriately cited when data are used for publication. The authors would appreciate learning about research projects, teaching exercises and other uses given to this data set and are open to contribute in further collaborations using this data.

6. Source funding

Fieldwork and specimen collection were carried out under multiple projects and funding sources throughout the years. We outline project names in italics and their funding sources in parenthesis.

Grupo de Monitoreo Ambiental (GEMA) and several other projects – (Instituto de Investigación de Recursos Biológicos Alexander von Humboldt, Ministerio de Ambiente

del Gobierno Nacional de Colombia), *Colección Teriológica CTUA* – (Universidad de Antioquia), *Colecciones de Ornitología y Mastozoología at Universidad Industrial de Santander, "Diversidad biológica en una región del sur de Bolívar, Colombia" (VIE-UIS 8867)* – (Ministerio de Ciencia, Tecnología e Innovación, Ministerio de Educación Nacional, Ministerio de Industria, Comercio y Turismo, and ICETEX, Ecosistema Científico-Colombia Científica by Fondo Francisco José de Caldas; Grant RC-FP44842-212-2018), *"Investigación de la biodiversidad y los servicios ecosistémicos para la gestión integral del territorio – descubriendo los ecosistemas estratégicos para el fortalecimiento de la gobernanza en el departamento de Santander (Santander-BIO)"* – (Sistema General de Regalias de la República de Colombia), *SELVA's Crossing the Caribbean and Neotropical Flyways projects* – (Cornell Lab of Ornithology, Rufford Small Grants, Environment and Climate change Canada, Guelph University, Acadia University), "*Influence of Climate on the Origin and Maintenance of Tropical Bird Communities"* (National Geographic Society Waitts Grant No. W270-13, Frank Champan fund from AMNH, Wilson Ornithological Society and Sigma Xi), *"Proyecto Gradientes"* (National Science Foundation Grant DEB-1120682, Universidad ICESI and EPSA-Celsia)*, Colección Zoológica* - (Universidad del Tolima, Programa Monitoreo de Aves en el bosque seco tropical del norte del Tolima, localidad Armero-Guayabal 2010-2019) – *HERC projects* (Rufford Small Grants, Vicerrectoría de Investigaciones y Posgrados, Universidad de Caldas, proyectos 0743919, 0406718 and 0277620), *Barreras de control peri-urbanas: el papel de los mamíferos silvestres en la circulación de hemoparásitos en la red Ecoparques de Manizales, Caldas* – (Vicerrectoría de Investigaciones y Posgrados Universidad de Caldas, Minciencias: no.112777758193 and 858-2017), "*Establecimiento de Áreas de Importancia para la Conservación de las Aves – AICAs – en el departamento de Putumayo. Fases II y III*" (Corporación para el desarrollo sostenible del sur de la Amazonia – CORPOAMAZONIA; proyectos I-06-086-1-02-04 10-12 y I-06-086-57302-04-05 53-13).

B. Specific subproject description

1. Site description

The Colombian territory is characterized by high topographical complexity that results in a wide range of environmental conditions which shape the distribution of its magnificent

biodiversity (Chapman 1917, IGAC 2008, Colombia, S. I. B. 2016). From the Caribbean basin (12˚31'N) to the Amazon river (04˚14'S), and from the Pacific Ocean (79˚13'W) to the Orinoco and Negro rivers (66˚51'W), continental Colombia can be divided into nine biogeographic regions (Hernández-Camacho *et al.* 1992, MinAmbiente 2017); all of these regions are represented in our data set. Specifically, our data set consists of body masses obtained from birds and mammals collected or captured and released throughout 2834 localities in Colombia (Figure 2). Some of the data are records from protected areas included in Colombia's national natural park system (PNN: Parques Nacionales Naturales). We also provide records from the Caribbean Archipelago of San Andrés and Providencia islands (12˚-14˚ N; 81˚40'W).

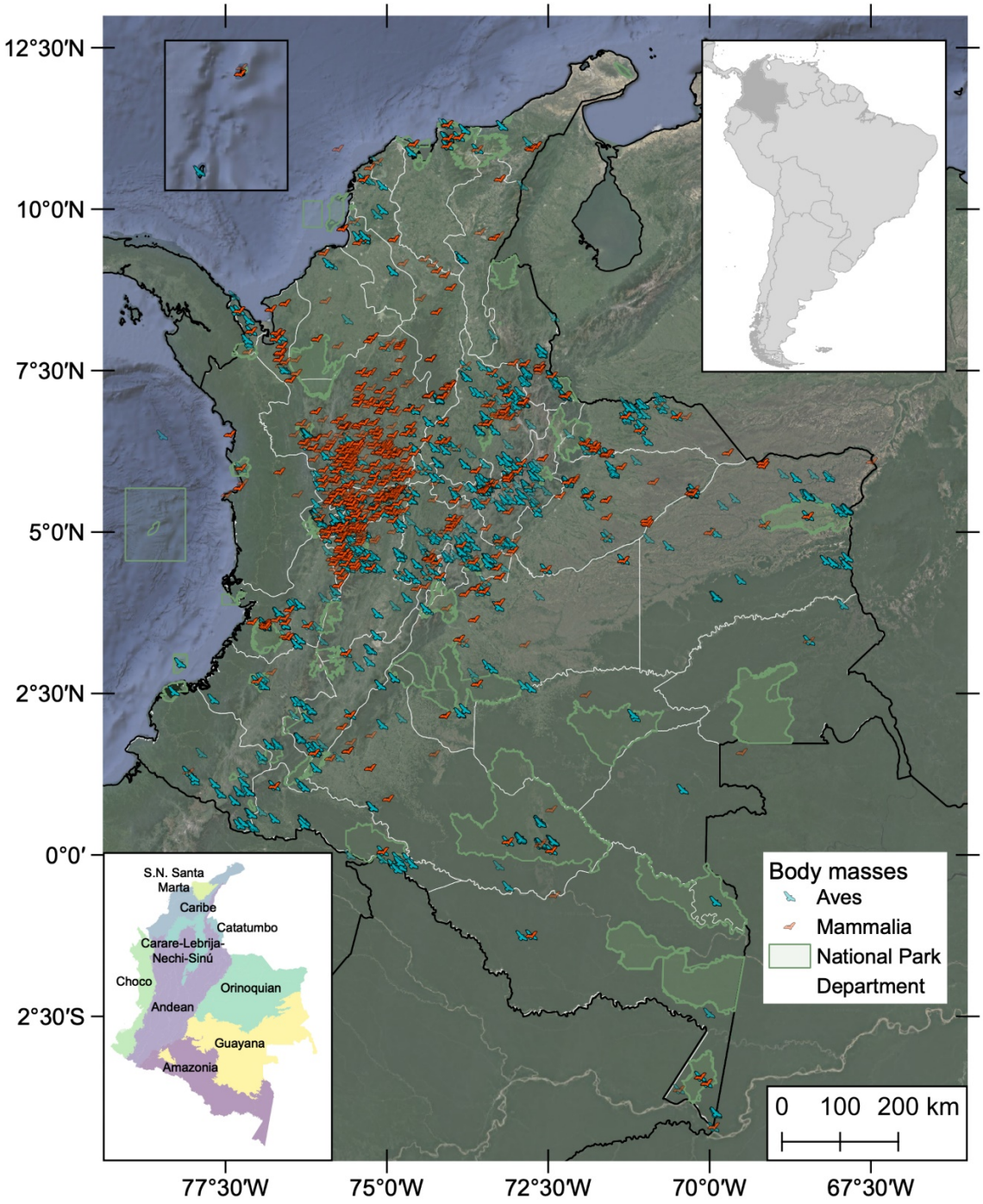

Fig. 2. Map showing localities where bird and mammal body mass was collected. Inset map on the top-right shows the location of Colombia within South America (in dark gray). Upper-left inset shows San Andrés and Providencia islands. The inset map on the lower-left shows the nine biogeographic regions currently recognized in Colombia (MinAmbiente 2017).

2. Data acquisition

We compiled body mass data from 19159 specimens housed in the following scientific collections: Colecciones Biológicas, Instituto de Investigación de Recursos Biológicos Alexander von Humboldt (IAvH-A & IAvH-M); Colección Teriológica, Universidad de Antioquia (CTUA); Colección de Ornitológia y Mastozología, Museo de Historia Natural, Universidad Industrial de Santander (UIS-MHN-O & UIS-MAH-M); Grupo de Investigación en Zoología, Universidad del Tolima (CZUT); Museo de Historia Natural, Universidad de Caldas (MHN-UCa); and data from 30304 individuals captured and measured at banding stations operated by: SELVA: Investigación para la Conservación en el Neotrópico; Departamento de Química y Biología, Universidad del Norte; Reserva Natural Los Yátaros; Departamento de Ciencias Biológicas, Universidad ICESI; Fundación Reserva Natural La Palmita, Grupo de investigaciones territoriales para el uso y conservación de la biodiversidad.

3. Taxonomy and data cleaning

For birds, we followed the taxonomy adopted by Asociación Colombiana de Ornitología (Avendaño *et al.* 2017, ACO 2020). It includes most changes suggested by the South American Classification Committee (Remsen *et al*. 2019) but some adjustments have been made (Acevedo-Charry *et al.* 2020). Mammal taxonomy follows the checklist of Colombian mammals (Ramírez-Chaves *et al.* 2019). We also adjusted the mammal checklist to reflect recent changes and to include recently described species such as *Anotomys leander* Thomas, 1906 (Marin-C and Sanchez-Giraldo 2017), *Gardnerycteris koepckeae* Gardner and Patton, 1972 (Morales-Martínez *et al*. 2020), *Eptesicus miradorensis* Allen, 1866 (Turmelle *et al.* 2011), *Makalata macrura* Wagner, 1842 (Emmons *et al.* 2015) and *Philander melanurus* Thomas, 1899 (Voss *et al*. 2018). *Artibeus* species follow Simmons and Cirranello (2020).

To account for measurement error or data set entry mistakes we identified outlier data for each species using the identify_outliers function in the rstatix package for R (R Core Team 2013, Kassambara 2020). Outliers were defined as individual observations larger or smaller than 1.5 times the interquartile distance of the body mass distribution for the species the outlier belonged to. Outliers were contrasted to the original data and

corrected when possible and removed in case the source of error was not identifiable. After data base cleaning, we removed a total of 2570 records from the data set.

All coordinates are provided in decimal degrees. When coordinates were from the original source in any other format they were standardized to decimal degrees using canadensys tool (https://www.canadensys.net). When standardization was applied the original sexagesimal information was kept in the verbatimCoordinate column. Records were validated using spatial Arcgis 10.2 analysis tools based on recent official cartography (IGAC 2019). For some observations with missing or mismatching coordinates but with a named locality was provided from the original source, we geo-referenced the locality using the SiB Colombia gazetteer (Colombia, S. I. B. 2016, https://sibcolombia.net). Georeferencing comments and remarks are detailed in the database in the georreferenceRemaks column.

4. Data description

We included body masses for 1317 bird species representing 29 taxonomic orders (Figure 3, Table 1), ranging from 1.90 gr (*Phaethornis ruber* and *Chlorostilbon mellisugus*, Trochilidae, Apodiformes) to 6700 gr (*Jabiru mycteria*, Ciconiidae, Ciconiiformes). We also included body masses from 270 mammal species representing 11 orders and ranging between 2.00 gr (*Rhogessa io*, Vespertilionidae, Chiroptera) and 11000 gr (*Cuniculus paca*, Cuniculidae, Rodentia). We present average body mass for every species in the data set, standard deviation (±SD) and sample size (n). We also separate the data among males, females and unknown sex individuals.

Table 1. Number of individuals and species for all bird and mammal orders reported in this data set.

Class	Order	No. individuals	No. species
Aves	Ciconiiformes	1	1
	Eurypygiformes	5	1
	Opisthocomiformes	6	1
	Procellariiformes	1	1
	Sphenisciformes	1	1
	Steatornithiformes	12	1
	Nyctibiiformes	6	3
	Podicipediformes	9	3
	Suliformes	14	3
	Cathartiformes	25	6
	Coraciiformes	228	11
	Cuculiformes	137	11
	Tinamiformes	48	11
	Falconiformes	65	12
	Trogoniformes	90	12
	Anseriformes	78	14
	Gruiformes	60	14
	Caprimulgiformes	93	15
	Pelecaniformes	50	15
	Strigiformes	83	16
	Columbiformes	289	26
	Galliformes	119	26
	Charadriiformes	78	29
	Accipitriformes	109	31
	Galbuliformes	277	31
	Psittaciformes	177	39
	Piciformes	455	54
	Apodiformes	5826	131
	Passeriformes	33680	798
Mammalia	Artiodactyla	1	1
	Cingulata	1	1
	Lagomorpha	1	1
	Pilosa	1	1
	Paucituberculata	28	2
	Primates	3	2
	Eulipotyphla	25	4
	Carnivora	14	8
	Didelphimorphia	188	19
	Rodentia	1471	78
	Chiroptera	5707	153

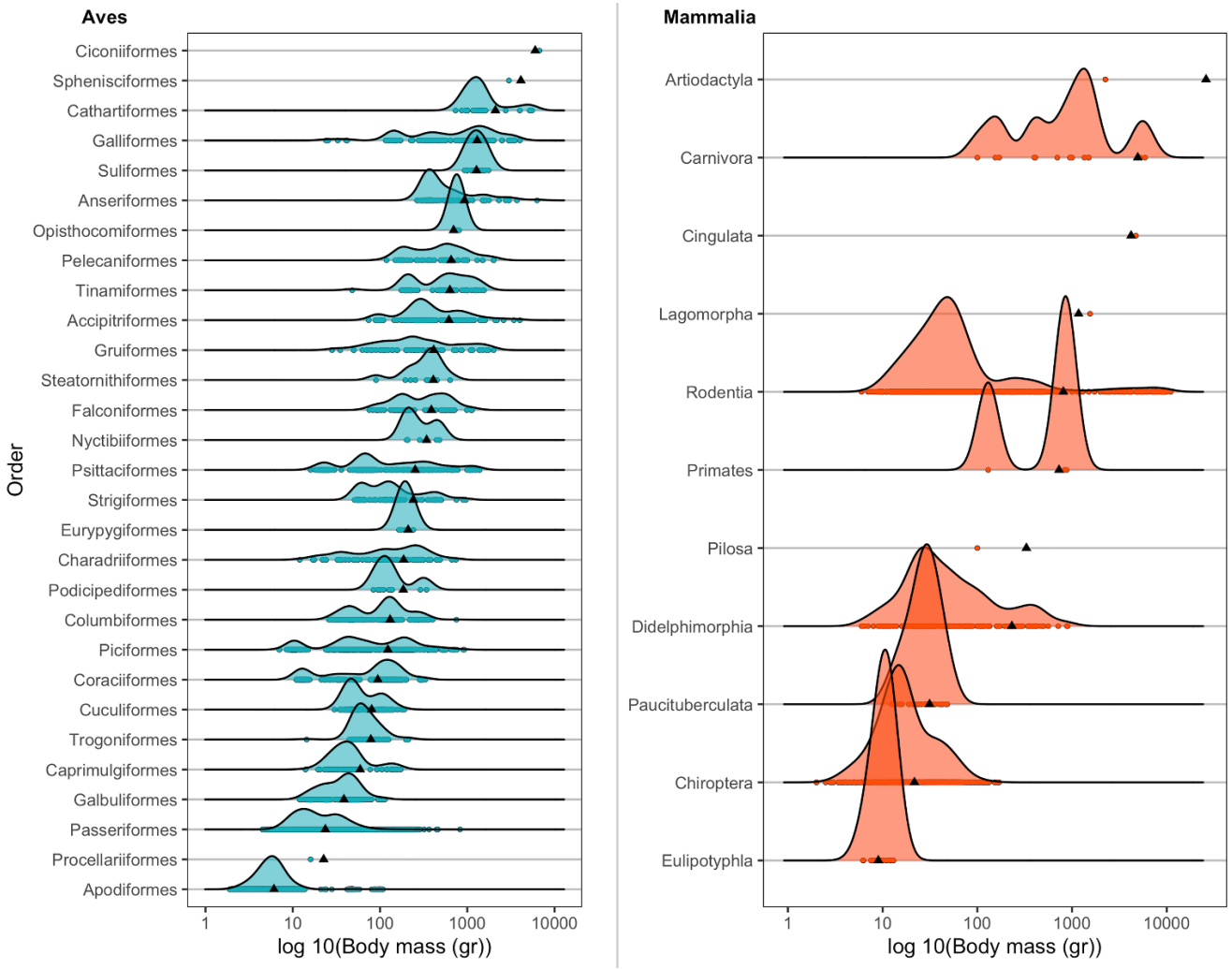

Fig. 3. Bird and mammal body mass density plot by order. Black triangles show the mean for each order reported in the EltonTrait data set (Willman et al. 2014).

5. **Project personnel**

David Ocampo[1], Kevin G. Borja-Acosta[1], Julián Lozano-Flórez[1], Sebastián Cifuentes-Acevedo[1,2], Enrique Arbeláez-Cortés[3], Nicholas J. Bayly[4], Ángela Caguazango[4], Brayan Coral-Jaramillo[5], Diego Cueva[1], Fernando Forero[1,6], Juan P. Gómez[7], Camila Gómez[4,8], Mario A. Loaiza-Muñoz[9], Gustavo A. Londoño[9], Sebastián Pérez-Peña[1], Sergio Losada-Prado[10], Héctor E. Ramírez-Chaves[11], Miguel E. Rodríguez-Posada[12], Jeyson Sanabria-Mejía[4], Manuel Sánchez-Martínez[9], Victor Hugo Serrano-Cardozo[13], M. del Socorro Sierra-Buitrago[1], Juliana Soto-Patiño[1] & Orlando Acevedo-Charry[1,14]
(see affiliations' above).

Class III. Data set status and accessibility

A. Status

1. **Latest update**

August 2020.

2. **Latest archive**

October 2020.

3. **Metadata status**

Complete and up to date on October 2020.

4. **Data verification**

Data set was checked and cross-validated by lead and senior authors (D. Ocampo and O. Acevedo-Charry) after first validation by every author before publication. Particular data-mining and multiple-checking were performed by K. G. Borja-Acosta, J. Lozano-Flórez and S. Cifuentes-Acevedo. Potential mistakes or absences should be reported to D. Ocampo or O. Acevedo-Charry.

B. Accessibility

1. **Storage location and medium**

The data are available as Supporting Information to this Data Paper publication in Ecology. A digital version of the data set is held by the authors. We provided links to data set in https://figshare.com; see Class V: Section B, Archiving.

2. **Contact person**

David Ocampo (docampo@humboldt.org.co) or Orlando Acevedo-Charry (oacevedo@humboldt.org.co, acevedocharry@gmail.com), Instituto de Investigación de Recursos Biológicos Alexander von Humboldt, Claustro de San Agustín, Villa de Leyva, Boyacá, Colombia.

3. Copyright restrictions

None

4. Proprietary restrictions

None. The present data paper should be appropriately cited when data are used for publication. We would appreciate learning about research projects, teaching exercises and other uses given to this data set. Authors are open to contribute in further collaborations using this data.

5. Cost

None.

Class IV. Data structural descriptors

A. Data set file

1. Identity

DataS1_Body mass by species_DOetal.csv

DataS1_Body mass records_DOetal.csv

2. Size

1588 species of 16 variables

49464 records of 24 variables

3. Format and storage mode

A single table available in excel file.

A single table available in excel file.

4. Header information

The first row of the file contains variable names.

Class V. Supplemental descriptors

A. Data acquisition

1. Data forms or acquisition methods

We compiled body mass data from 19159 specimens housed in the following scientific collections: Colecciones Biológicas, Instituto de Investigación de Recursos Biológicos Alexander von Humboldt (IAvH-A & IAvH-M); Colección Teriológica, Universidad de Antioquia (CTUA); Colección de Ornitológia y Mastozología, Museo de Historia Natural, Universidad Industrial de Santander (UIS-MHN-O & UIS-MAH-M); Grupo de Investigación en Zoología, Universidad del Tolima (CZUT); Museo de Historia Natural, Universidad de Caldas (MHN-UCa); and data from 30304 individuals captured and measured at banding stations operated by: SELVA: Investigación para la Conservación en el Neotrópico; Departamento de Química y Biología, Universidad del Norte; Reserva Natural Los Yátaros; Departamento de Ciencias Biológicas, Universidad ICESI; Fundación Reserva Natural La Palmita, Grupo de investigaciones territoriales para el uso y conservación de la biodiversidad.

The data sets are presented in the supplementary material as follow: DataS1**_Body mass by species**, data set of average body mass for every species, standard deviation (±SD) and sample size (n). We also separate the data among males, females and unknown sex individuals. And DataS1**_Body mass records**, data set with all the single records including metadata with information about record type (i.e. Preserved specimen or Capture-release), taxonomy, ID, locality, date and original resource.

2. Quality assurance/quantity control procedures

To account for measurement error or data set entry mistakes we identified outlier data for each species using the identify_outliers function in the rstatix package for R (R Core Team 2013, Kassambara 2020). Outliers were defined as individual observations larger or smaller than 1.5 times the interquartile distance of the body mass distribution for the species the outlier belonged to. Outliers were contrasted to the original data and corrected when possible and removed in case the source of error was not identifiable. After data base cleaning, we removed a total of 2570 records from the data set.

All coordinates are provided in decimal degrees. When coordinates were from the original source in any other format they were standardized to decimal degrees using canadensys tool (https://www.canadensys.net). When standardization was applied the original sexagesimal information was kept in the verbatimCoordinate column. Records were validated using spatial Arcgis 10.2 analysis tools based on recent official

cartography (IGAC 2019). For some observations with missing or mismatching coordinates but with a named locality was provided from the original source, we geo-referenced the locality using the SiB Colombia gazetteer (Colombia, S. I. B. 2016, https://sibcolombia.net). Georeferencing comments and remarks are detailed in the database in the georreferenceRemaks column.

B. Archiving

1. Redundant archival sites

Body mass by species_DOetal.xlsx and Body mass records_DOetal.xlsx are also posted on Figshare (Ocampo 2020): https://doi.org/10.6084/m9.figshare.13473147.v1

ACKNOWLEDGMENTS

We thank biological collections' staff and researchers working on specimen collection throughout all these years. To Mauricio Álvarez for leading the Grupo de Exploración y Monitoreo Ambiental (GEMA) at IAvH-A and IAvH-M collections. To all the field assistants involved in every project and banding station. NB and CGM to Claudia Weber – Hacienda La Victoria, Fundación Iguaraya – Reserva El Pantano y Refugio Guajiro, Ios Vieira – Reserva Quebrada Valencia, Fundación ProAves – Reserva El Dorado. GAL to Parques Nacionales Naturales de Colombia, for allowing us to work at Tatamá and Farallones de Cali National Natural Parks. JPG to Hda. Los Limones, Reserva Natural Bateas, Reserva Mana Dulce, H. Llara, Reserva Jabirú, Reserva Natural Rio Manso, Reserva Natural Rio Claro, Corantioquia, A. Link, Reserva Natural Hda. El Triunfo and G. Campuzano for allowing us to work on their properties.

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